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Analog Devices, Inc.

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A2B PLUG AND PLAY RELEASE NOTES

Document Status:	Approved
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Revision List**Table 1: Revision List**

Document Revision	Date	Description
0.1	18-11-2022	Updated for Plug and Play Rel 1.0.0
1.0	21-11-2022	Approved and Baselined for PnP 1.0.0
1.1	08-06-2023	Updated for Plug and Play Rel 1.1.0
2.0	09-06-2023	Approved and baselined for PnP 1.1.0
3.0	13-10-2023	Baselined for PnP 1.3.0

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1 Introduction

Enhancing A2B technology with Plug and Play (PnP) capability improves technology adoption in dynamic applications. PnP allows for seamless system integration in scenarios that include: on-stage musical connectivity, conferencing, and audio distribution within fixed installations. Plug and Play is helpful even in static configuration applications; it reduces configuration, testing & maintenance times.

1.1 Purpose

Software package contains Plug and Play Software which runs on top of A2B Stack. All the features provided in current version of Software are listed in this document.

1.2 Scope

This package has three major components.

- PnP software module along with A2B Stack in Source form or Library form
- A Windows terminal application for quick evaluation of PnP
- Embedded demo examples on ADSP-SC594 using EV-SC594-SOM

The core component of the package, which is 'PnP software module' is platform agnostic and can be ported to any other platform. Refer to **A2B_PnP_QuickStartGuide.pdf** for a jump start.

1.3 Organization of the document

Section 1 to 7 details about the content of the releases, the changes or the features which got added and other known issues/ problems in the release.

2 Release Information

2.1 Release Contents

Table 2: Release Contents

Sl. No	Release Item	Description	Details
1	A2B Plug and Play Software (<i>Source code</i>) A2B Stack Target Software (<i>source code</i>)	Development Release	Version 1.3.0
		Supported Hardware platform	A2B Evaluation Boards EVAL – AD2437A1NZ Rev D (Main) EVAL – AD2437B1NZ Rev D (Sub) EVAL – AD2437A1MZ Rev D(Main) EVAL – AD2437B1MZ Rev D (Sub) EV-SC594-SOM Rev B EV-SOMCRR-EZKIT Rev A
		Supported AD2437 Silicon revision	AD2437: R 2.1, R1.3
		Supported SigmaStudio version	SigmaStudio+ Version 2.1.0 A2B Plugin for SigmaStudio+ Version 1.3.0

3 Supported Features

3.1 Rel 1.3.0

Table 3: Supported Features

Release Number	Release Date	Features Supported
1.3.0	15 - 10 - 2023	• A Windows/PC terminal application to evaluate PnP (without the need of any embedded platform) • Multi -Main support • BW estimation, remote upgrade of PnP content, Support for 8-byte Vendor ID • Support for batch execution of application commands

3.2 Rel 1.1.0

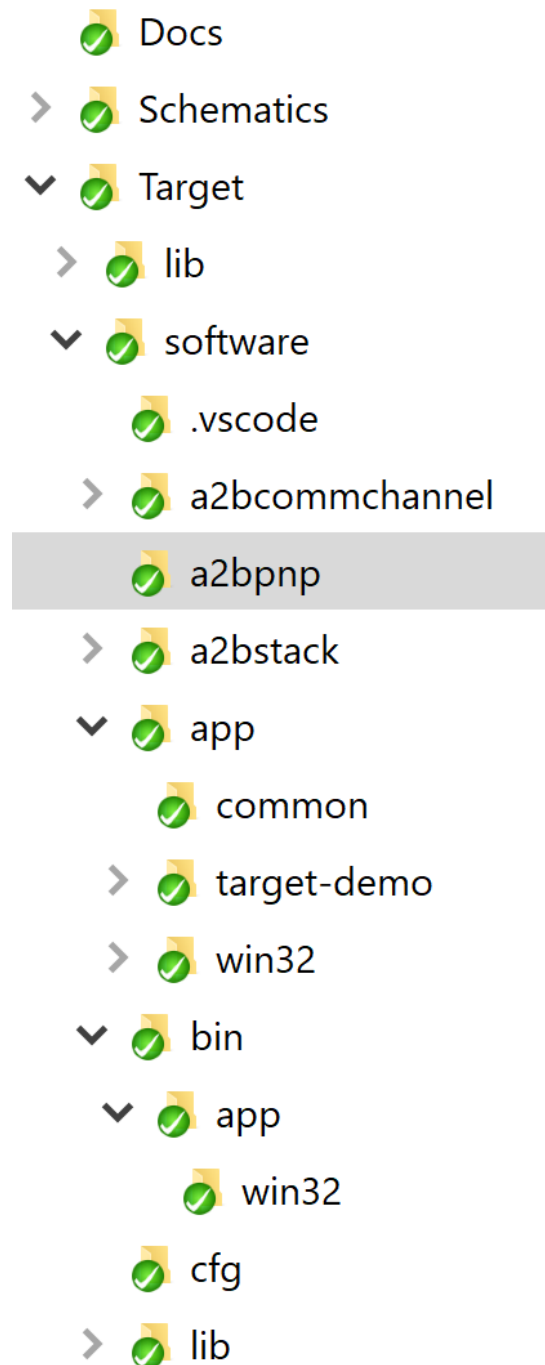
Table 4: Supported Features

Release Number	Release Date	Features Supported
1.1.0	09 - 06 - 2023	• A2B-SS+ plugin update based on updated PnP Spec • Added support for multiple peripherals in PnP Node. • Group based multiple peripheral Audio Routing. • Added Data Tunnel Support • APIs are added for GPIO over Distance • APIs are added for supplier information read • Tested audio routing involving main node audio peripherals. • Enhanced UART application for quick debug and evaluation

4 Package Details

The release package contains folder structure as shown below.

ADI_A2B_PnP_Software-Rel1.3.0



The below section explains the different folders and their purpose in the current release.

Table 5: Package Details

Folder Name	Purpose
Target/software	PnP module, A2B stack and examples
Target/software/bin/app/win32	This folder contains windows terminal app
Schematics	This folder contains the example A2B and SigmaDSP schematics (SS+)
Docs	This folder contains all the documents including quick start guide.

5 Package Installation

5.1 Windows

Double click the A2B Software package (executable) to install. The package is installed into
"C:\Analog Devices\ADI_A2B_PnP_Software -RelX.Y.Z"

6 Known Issues and Workarounds

6.1 Limitations

The following are some of the important Issues/limitations known at the time of this release:

- New node addition or deletion may momentarily mute the existing audio streams. Application is notified with impending mute for necessary DSP handling.
- Occasionally, plugging of the cable generates Indeterminate Bus Fault (interrupt type 0xF) interrupt. Currently, application ignores this interrupt and continues with discovery.
- Undocumented interrupt (0x8) is observed during the blind discovery. Currently, PnP ignores the interrupt, and it doesn't have any functional implication. Work is in progress to address this scenario.
- Unknown cable length detection during PnP is limited to 10m only. Work is in progress to support up to 30m.
- Windows application takes around 3-4s to respond to Bus drop, this due to firmware limitation within USBi adapter.

7 Documentation

- Refer to '**A2B_PnP_QuickStartGuide.pdf**' document to quickly set up an example demo application.
- Refer to '**A2B_PnP_UserGuide.pdf**' for detailed understanding of porting and application integration aspects.
- Refer to '**A2B_PnP_API_Reference.chm**' for API description and parameter details.
- Refer to '**A2B_PnP_ContentSpec.pdf**' to get an overview of Plug and Play content that need to be written to the device's EEPROM to make it compliant with ADI's Plug and Play Software.

8 Technical Support

8.1 Contact information

If you have a technical problem and you can't find a solution, you can contact for Technical Support at:

<mailto:a2bPnPSupport@analog.com>

8.2 Type of support

All technical queries, bug reporting, issues and feedbacks addressed to the above-mentioned contact shall be processed and responded accordingly based on the nature of the support required.

Terminology

Table 6: Terminology

Term	Description
PnP	Plug and Play
GUI	Graphical User Interface